

Acquisition Time in Dual Shunt-Based Current Sensing for PMSM FOC

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Abstract—High-efficiency field-oriented control (FOC) of permanent-magnet synchronous motors (PMSM) hinges on precise, low-latency stator current measurement synchronized to PWM. In sensorless drives, current-measurement error propagates through the position estimator and can degrade efficiency compared to sensed FOC. This article translates estimator accuracy targets into concrete ADC and PWM requirements, with a practical focus on shunt topologies. We show why acquisition time (settling + sample-and-hold time) is usually more critical than nominal analog-to-digital converter (ADC) bandwidth, how modulation index and space-vector modulation (SVM) windows constrain sampling, and how topology choices affect both precision and timing.

Index Terms—Embedded Systems, Motor Control, PMSM, Current sensing, Dual shunt

I. OVERVIEW AND SCOPE

Field-oriented control (FOC) of permanent-magnet synchronous motors (PMSMs) relies on precise, low-latency phase-current measurements that are time-aligned to the pulse-width modulation (PWM) waveform [1], [2]. While many analyses treat current measurement as ideal, practical inverter and signal-chain constraints (deadtime, switching transients, amplifier settling, ADC acquisition windows) make the *timing of acquisition* as important as nominal ADC resolution [3], [4]. This paper focuses on shunt-based current sensing and shows how current-measurement precision—especially in *sensorless* operation—drives position-estimator accuracy, efficiency performance [5]–[7].

A. Topologies considered

We review and compare four shunt topologies that dominate cost- and size-constrained drives visualised on figure 1:

- 1) **In-line phase shunts** (one per phase at the motor terminals): highest accuracy and simplest reconstruction. Requires isolation at higher bus voltages and careful handling of common-mode transients [3].
- 2) **Low-side three-shunt** (one shunt per phase return): high accuracy with reduced common-mode stress [1].
- 3) **Low-side two-shunt**: similar to three-shunt but lower bill of materials. The 3 shunt can be reconstructed still relatively simple with Kirchhoff's Current Law. $i_a + i_b + i_c = 0$ [4].

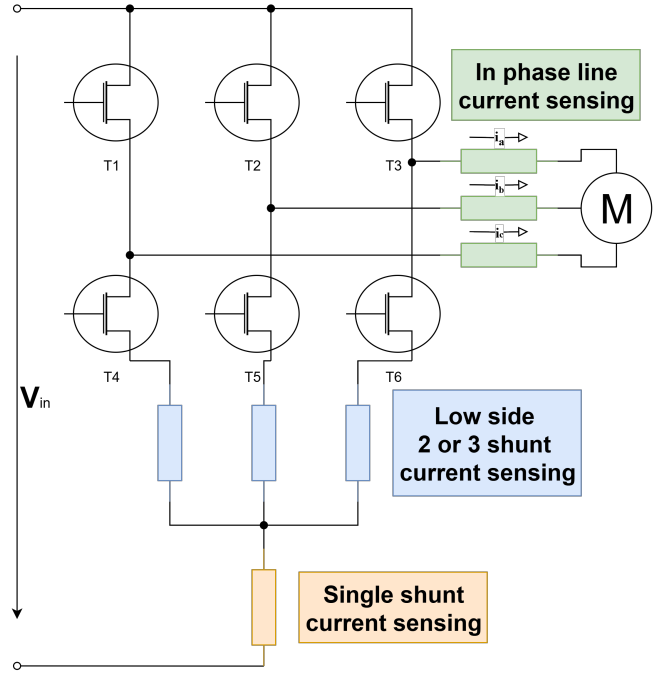


Fig. 1. 3-phase Current sensing topologies

- 4) **Low-side single-shunt** (DC return): most economical and compact, but requires current reconstruction and imposes the tightest sampling-window constraints at certain modulation indexes and sector boundaries [3], [8].

B. Control, PWM, and timing preconditions

We take as a *hard requirement* that the analog sample is fully acquired and settled *within* the valid PWM sampling window. If we don't wait for the acquisition time, the ADC's sample-and-hold latches a still-settling (wrong) value and introduce a hardly predictable value. [9]

In this article, further assumptions and scenarios are used to keep the equations clear.

- center-aligned PWM with space-vector modulation (linear SVM, split zero vectors);

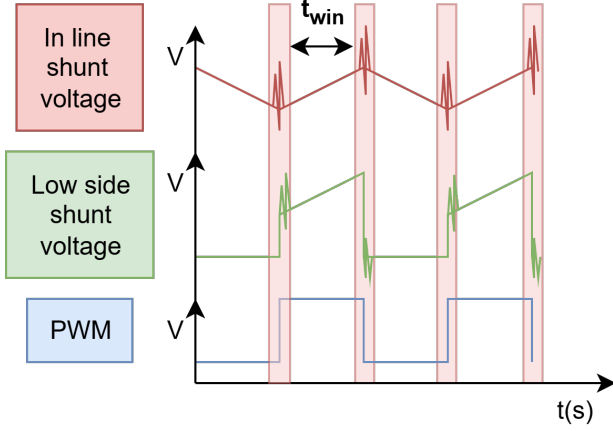


Fig. 2. Sampling window

- ADC sampling is triggered at the center of active-vector plateaus;
- no clamped SVM and no enforced minimum on-time unless stated otherwise;
- deadtime, gate-driver propagation delay, and switching transient decay are captured in t_{blank} ;
- trigger jitter and timer/ADC dispersion are captured in t_{margin} ;
- $T_{\text{sw}} = 1/f_{\text{sw}}$ denotes the full PWM period (carrier period).

$$t_{\text{acq}} \leq t_{\text{win}} \quad \text{with} \quad t_{\text{win}} = t_{\text{on}} - t_{\text{blank}} - t_{\text{margin}} \quad (1)$$

Meaning of terms:

- t_{acq} — **acquisition/settling time**: the analog time needed for the shunt $\rightarrow$ PGA/op-amp $\rightarrow$ ADC input (and sample/hold) to settle to the target error band.
- t_{win} — **valid sampling window**: the portion of the SVM segment where switching transients are sufficiently decayed to allow accurate sampling.
- t_{on} — **active-vector on-time** for the relevant SVM segment (sets the geometric “ceiling” of the window).
- t_{blank} — **blanking time** after edges (deadtime + edge ringing / common-mode feedthrough and any enforced hardware blanking).
- t_{margin} — **guard time** for clock jitter, trigger dispersion.

The (1) is a *physical constraint* for FOC accuracy. By contrast, many works emphasize the ADC “throughput” interval $t_{\text{acq}} + t_{\text{conv}}$ or headline mega sample per second (MSPS) and treat it as the primary selection metric; however, once the sample is captured and held *inside* t_{win} , the conversion time t_{conv} may complete later (even in the next half-period) without degrading the instantaneous measurement fidelity [3], [4]. Consequently, MSPS is not a reliable technical metric

for direct evaluation, unless equation (1) is satisfied. See representative throughput-centric treatments in [10]–[15].

For dual and triple shunt measurements we assume space-vector modulation (SVM) with center-aligned PWM, and schedule ADC triggers at the *centers of active vectors* to minimize switching transients [3], [8]. The blanking term t_{blank} includes dead-time and edge-ringing/CM feedthrough; t_{margin} covers clock jitter and trigger dispersion.

ADC throughput $(t_{\text{acq}} + t_{\text{conv}})^{-1}$ still matters for picking the *loop update rate* (single or double update per carrier), DMA bandwidth, and multi-channel interleaving, but it is *secondary* to (1) for measurement accuracy. Meeting (1) enables deterministic, center-of-vector sampling, after which conversion may pipeline into the next half-period without penalty [16], [17].

C. Precision challenge and why acquisition time matters

In sensorless FOC, the position estimator maps current error to electrical angle error; small increases in current-measurement noise/offset directly increase torque ripple and copper loss, particularly at low speed/weak back-EMF [18]. Achieving the required *system* Effective number of bits (ENOB) at the control update rate therefore hinges on capturing a clean sample *inside* t_{win} . Headline ADC bandwidth or “MSPS” figures are insufficient if the input network cannot settle within the available window [9]. Throughout the paper we emphasize specifying and validating *acquisition/settling time* rather than generic bandwidth.

D. Modulation index definition (for clarity)

To avoid ambiguity, we define the SVM modulation index as

$$m = \frac{V_{\text{ref}}}{V_{\text{dc}}/\sqrt{3}}, \quad (2)$$

so that $m=1$ corresponds to the linear SVM limit [3], [8]. Alternative definitions exist (e.g., based on line-to-line peak). There are other modulation index definitions easy to mix. Discussions of single-shunt blind spots, minimum on-time, and clamped SVM strategies in this article assume (2).

E. Performance metrics used in this paper

We evaluate topologies using:

- **System ENOB at loop bandwidth** (includes sensor, PGA, ADC) [4].
- **Acquisition/settling time** to a specified error band (e.g., $\pm \frac{1}{2}$ LSB or ppm target) under realistic common-mode steps [4].
- **Window robustness** vs. modulation index m (presence/absence of blind spots; need for clamped SVM or minimum on-time) [3], [8].

II. FROM CURRENT PRECISION TO SENSORLESS ESTIMATOR ACCURACY AND EFFICIENCY

In sensorless FOC, the rotor electrical angle θ_e is inferred from the applied voltages and measured currents through an observer or phase-locked loop (PLL) based on the PMSM model [2], [6], [19]. Any error in the measured phase currents $\{i_a, i_b, i_c\}$ propagates to the dq currents, back-EMF estimates, and ultimately to the angle estimate.

A. Measurement error taxonomy

We decompose the instantaneous current error (referred to the ADC input after gain) as

$$\epsilon_I^2 = \epsilon_q^2 + \epsilon_{\text{amp}}^2 + \epsilon_{\text{EMI}}^2 + \epsilon_{\text{jitt}}^2 + \epsilon_{\text{recon}}^2 + \epsilon_{\text{off}}^2, \quad (3)$$

where ϵ_q is quantization noise, ϵ_{amp} amplifier/programmable gain amplifier (PGA)/anti-alias noise, ϵ_{EMI} PWM-correlated interference not fully blanked, ϵ_{jitt} aperture-jitter induced error, ϵ_{recon} reconstruction/phase-skew error (notably for two-shunt and single-shunt), and ϵ_{off} residual offset after calibration.

B. Torque, ripple, and efficiency impact

For a surface-PM motor with direct-axis current $i_d \approx 0$, the torque is $T = \frac{3}{2}p\lambda i_q$ [1]. λ is flux linkage motor specific constant and p is the number of pole pairs. With an electrical angle error $\Delta\theta$, the dq frame is rotated and the effective torque becomes

$$T' = \frac{3}{2}p\lambda (i_q \cos \Delta\theta + i_d \sin \Delta\theta) \approx T \cos \Delta\theta, \quad (4)$$

assuming $i_d = 0$. To hold the same torque, the controller increases $|i_q|$ by $1/\cos \Delta\theta$. Copper loss scales with current squared, so the relative increase is

$$\frac{\Delta P_{\text{cu}}}{P_{\text{cu}}} \approx \frac{1}{\cos^2 \Delta\theta} - 1 \approx \Delta\theta^2 \quad (\Delta\theta \text{ rad, small-angle}) \quad (5)$$

which shows why even small RMS angle errors (e.g., 1° – 2°) can measurably affect efficiency.

C. Setting current and ADC precision targets

Let the desired RMS angle error be ϵ_θ (e.g., 1° – 2°), and let $S_{\theta I}^{\text{max}}$ be the worst-case sensitivity over the operating region of interest. Then the allowable RMS current error is

$$\epsilon_I = \frac{\epsilon_{\theta, \text{tar}}}{S_{\theta I}^{\text{max}}}. \quad (6)$$

Allocate a fraction β (e.g., 0.3) of this budget to quantization, $\epsilon_q = \beta \epsilon_{I, \text{tar}}$. With ADC full-scale current I_{FS} (after gain), quantization RMS is $\epsilon_q = I_{FS}/(2^N \sqrt{12})$. Solving for the required *effective* number of bits at the control update rate:

$$N_{\text{ENOB}} \geq \log_2 \left(\frac{I_{FS}}{\sqrt{12} \beta \epsilon_I} \right). \quad (7)$$

Using a crest factor $C = I_{FS}/I_r$ (often 1.5–3) and a relative current error ε ($\sigma_{I, \text{tar}} = \varepsilon I_r$) gives the practical form [20]

$$N_{\text{ENOB}} \geq \log_2 \left(\frac{C}{\sqrt{12} \beta \varepsilon} \right), \quad (8)$$



Fig. 3. Analogue signal chain, each element adding time delay to the measured signal until the control loop processes

with additional margin reserved for amplifier/EMI/reconstruction/offset terms [4].

D. Acquisition time, settling, and windowing

To realize the precision implied by (8) the analog front-end (shunt + PGA/op-amp + ADC input network) must settle inside the valid PWM window. Signal chain showed on figure. 3

E. Sensorless vs. sensed baseline

In **sensed** FOC (encoder/resolver), the rotor electrical angle is *measured*, so the alignment error is effectively zero ($\Delta\theta \approx 0$). With no angle penalty, the torque expression reduces to

$$T' \approx \frac{3}{2}p\lambda (i_q + e_{i_q}), \quad (9)$$

so the only path from measurement to torque error is the *current-measurement error* e_{i_q} itself (plus any inverter ripple), not a compounded $\cos \Delta\theta$ loss 4. Practically, this means sensed FOC can tolerate slightly lower current-measurement precision for the same torque ripple/efficiency, because there is no misalignment term multiplying the error [2].

Let $r \triangleq \epsilon_I/I_{q, \text{rms}}$ be the relative RMS current-measurement error. Copper loss scales with current squared, so:

$$\left. \frac{\Delta P_{\text{cu}}}{P_{\text{cu}}} \right|_{\text{sensed}} \approx r^2. \quad (10)$$

In **sensorless** FOC, current-measurement error ϵ_I also drives *angle* error via the estimator sensitivity $S_{\theta I}$, giving an RMS angle error $\epsilon_\theta \approx S_{\theta I} \epsilon_I$. The misalignment requires extra i_q by $1/\cos \Delta\theta$, so from (5) the copper-loss increase adds $\Delta\theta^2$ (small-angle). Hence

$$\left. \frac{\Delta P_{\text{cu}}}{P_{\text{cu}}} \right|_{\text{sensorless}} \approx r^2 + \epsilon_\theta^2 = r^2 + (S_{\theta I} \epsilon_I)^2. \quad (11)$$

To map copper-loss growth to *efficiency*, let η_0 be the baseline efficiency and $\kappa \triangleq P_{\text{cu}}/P_{\text{in}}$ the baseline copper-loss share of input power at the operating point. Holding output power constant,

$$\eta_{\text{sensed}} \approx \frac{\eta_0}{1 + \kappa r^2}, \quad \eta_{\text{sensorless}} \approx \frac{\eta_0}{1 + \kappa (r^2 + \epsilon_\theta^2)}. \quad (12)$$

Thus, even with identical current-measurement quality (r), sensorless control pays an *extra* efficiency penalty proportional to ϵ_θ^2 , which is why ϵ_I must meet the target from (6) to bound ϵ_θ (and efficiency) [6], [19].

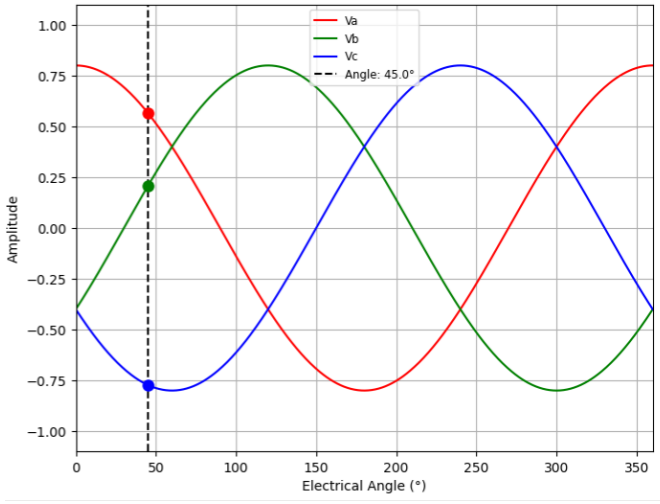


Fig. 4. Electrical position and modulation index ($m = 0.80$) defines duty cycles.

III. SAMPLING WINDOW, ACQUISITION TIME, AND MODULATION BOUNDARIES

With ENOB *assumed adequate* (Sec. II), the dominant constraint for accurate current measurement is whether the analog front end can *settle* within the valid sampling window created by the PWM/SVM pattern.

A. Definitions and precondition

We adopt the precondition from (1):

$$t_{\text{acq}} \leq t_{\text{win}} \quad (13)$$

Here t_{acq} is the analog acquisition/settling time to the target error band. t_{blank} covers edge-related blanking (deadtime + common-mode/ringing decay), and t_{margin} guards against jitter/trigger dispersion. It is convenient to group the requirements for minimum t_{win} as t_{req} required sampling time.

$$t_{\text{req}} \triangleq t_{\text{acq}} + t_{\text{blank}} + t_{\text{margin}}. \quad (14)$$

A sample inside a given active vector is valid if its on-time satisfies $t_{\text{on}} \geq t_{\text{req}}$. The conversion time t_{conv} is *not* rate-limiting provided the sample-and-hold has captured the settled value inside t_{win} [3], [4].

B. Valid sampling window for dual shunt

Unlike the single-shunt case, *dual-shunt* sensing does not impose a modulation *lower* bound from settling-time/overlap constraints, because the two phase currents can be sampled at *two distinct instants* within the half-carrier when each measured phase has its own valid plateau. The only hard timing constraint is an **upper** bound on modulation index m to keep each measured phase's plateau long enough to satisfy the required acquisition/settling time $t_{\text{req}} = t_{\text{acq}} + t_{\text{blank}} + t_{\text{margin}}$.

For center-aligned sinusoidal/SVM modulation, each phase duty is

$$d_a = \frac{1}{2} + \frac{m}{2} \sin \theta, \quad (15)$$

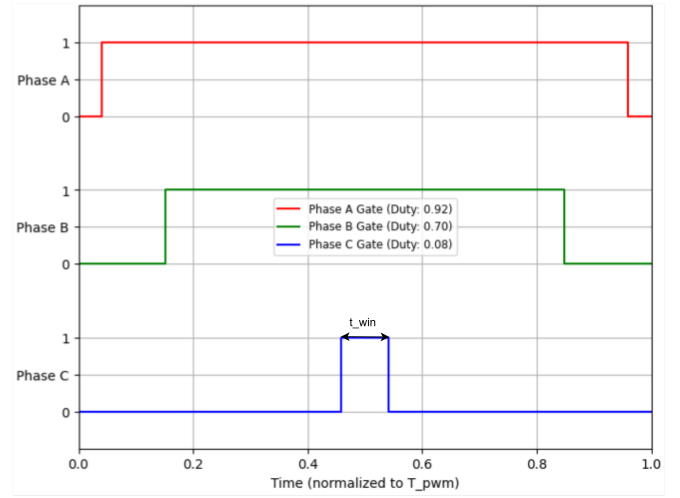


Fig. 5. Duty cycle in Phase C is getting to the sampling window limitation ragne.

$$d_b = \frac{1}{2} + \frac{m}{2} \sin(\theta - \frac{2\pi}{3}), \quad (16)$$

$$d_c = \frac{1}{2} + \frac{m}{2} \sin(\theta + \frac{2\pi}{3}), \quad (17)$$

with electrical angle θ and modulation index m as in (2). A phase's *shorter* plateau per carrier (low or high, depending on current direction and gating) has length

$$T_{\text{plat},k}^{\min} = \min(d_k, 1 - d_k) T_{sw} \geq \left(\frac{1}{2} - \frac{m}{2}\right) T_{sw} \quad (18)$$

- worst case over θ - so guaranteeing a valid sampling window for each measured phase $k \in \{i, j\}$ requires

$$\left(\frac{1}{2} - \frac{m}{2}\right) T_{sw} \geq t_{\text{req}} \implies m \leq 1 - \frac{2 t_{\text{req}}}{T_{sw}}. \quad (19)$$

Design takeaway: For dual-shunt, settling-time constraints create *no lower bound* on m ; the *only* hard bound from t_{req} is the *upper* limit (19). Reduce t_{acq} (and thus t_{req}) to extend usable m toward 1 without PWM shaping. Otherwise enforce a minimum on-time to guarantee per-phase plateaus at the expense of modulation headroom [21].

If a minimum on-time (t_{min}) or clamped SVM is enforced, replace t_{req} by $\max(t_{\text{req}}, t_{\text{min}})$, which tightens the admissible modulation headroom accordingly.

C. Single-shunt sampling window — key challenges

In single-shunt sensing the valid sampling window exists only within the active-vector plateaus of SVM and *shrinks* near sector edges; two clean samples per half-carrier are required, so losing either window breaks reconstruction. The dominant challenges are [22]:

- **Window dependence on operating point:** Valid windows vary with modulation index and sector angle; “edge” blind zones appear where one dwell time collapses.

- **Acquisition-time budget:** The required time $t_{\text{req}} = t_{\text{acq}} + t_{\text{blank}} + t_{\text{margin}}$ must fit inside each window; larger t_{req} reduces angle coverage and tightens modulation headroom.
- **Trigger determinism & jitter:** Samples must be placed near vector centers with tight timing; jitter directly converts to current error.
- **PWM shaping trade-offs:** Clamped SVM or enforced minimum on-time can guarantee windows at all angles, but they consume zero-vector dwell and reduce effective voltage (modulation) headroom.
- **Switching-frequency trade-offs:** Lower f_{sw} enlarges windows but increases current ripple/EMI; higher f_{sw} does the opposite.
- **Analog front-end realism:** Op-amp/PGA common-mode step response, layout/CMRR, and input RC dominate t_{acq} . These effects must be validated on hardware, not inferred from MSPS alone.

These challenges govern feasibility once ENOB is adequate; detailed formulas and mitigation strategies (coverage, bounds, and shaping) are deferred to a follow-up paper.

D. Practical example (dual-shunt, 150 ns ADC + 600 ns op-amp)

Assume $t_{\text{acq}} = 0.75 \mu\text{s}$ composed of **150 ns** ADC settling and **600 ns** op-amp/PGA settling (e.g., DRV8305 at gain 20) [13]. In dual-shunt, settling time imposes *no lower bound* on modulation; the hard constraint is the *upper* bound ensuring each measured phase has a plateau $\geq t_{\text{req}} = t_{\text{acq}} + t_{\text{blank}} + t_{\text{margin}}$:

$$\left(\frac{1}{2} - m\right)T_{\text{sw}} \geq t_{\text{req}} \Rightarrow \boxed{m \leq 1 - \frac{2t_{\text{req}}}{T_{\text{sw}}}}.$$

Best-case (minimal guard).: Let $t_{\text{blank}} + t_{\text{margin}} \approx 0 \Rightarrow t_{\text{req}} = 0.75 \mu\text{s}$. At $f_{\text{sw}} = 20 \text{ kHz}$ ($T_{\text{sw}} = 50 \mu\text{s}$):

$$m_{\text{max}} = 1 - \frac{2 \cdot 0.75}{50} \approx \mathbf{0.97}. \quad (20)$$

Interpretation: you can run essentially up to the linear SVM limit; only the top $\sim 3\%$ of modulation is excluded by settling.

Realistic guard.: Let $t_{\text{blank}} + t_{\text{margin}} = 0.30 \mu\text{s} \Rightarrow t_{\text{req}} = 1.05 \mu\text{s}$. Then

$$m_{\text{max}} = 1 - \frac{2 \cdot 1.05}{50} \approx \mathbf{0.958}. \quad (21)$$

Interpretation: with modest blanking/jitter margin, dual-shunt comfortably supports $m \lesssim 0.96$ at 20 kHz; operation at $m = 0.95$ is fully compatible without PWM shaping.

$$\begin{aligned} t_{\text{req}} = 0.75 \mu\text{s} : d_{\text{max}} &= 1 - \frac{0.75}{50} = 0.985 \quad d_{\text{min}} = 1.5\% \\ t_{\text{req}} = 1.05 \mu\text{s} : d_{\text{max}} &= 1 - \frac{1.05}{50} = 0.979, \quad d_{\text{min}} = 2.1\%. \end{aligned} \quad (22)$$

With the 150 ns ADC + 600 ns op-amp settling times, dual-shunt supports per-phase duties up to $\approx 98\%$ (best case) or $\approx 98\%$ – 97.9% with realistic margins, while maintaining the required settling window on every measured phase.

Section 3 — Summary

With ENOB already adequate, accurate current sampling hinges on the analog chain settling *inside* the PWM window; define the required window as $t_{\text{req}} = t_{\text{acq}} + t_{\text{blank}} + t_{\text{margin}}$, and note that conversion time is not limiting once the S&H captures a settled value. For **dual-shunt**, settling time imposes only an *upper* modulation bound $m \leq 1 - \frac{2t_{\text{req}}}{T_{\text{sw}}}$, which also caps the per-phase duty cycle to $d_{\text{max}} = 1 - \frac{t_{\text{req}}}{T_{\text{sw}}}$, $d_{\text{min}} = \frac{t_{\text{req}}}{T_{\text{sw}}}$. The equation is depending only on PWM frequency and analogue signal chain, but not on ADC bandwidth (t_{conv} does not apply).

IV. CONCLUSION

This work reframed dual shunt current sensing for PMSM FOC around *timing geometry*. Once the system ENOB is adequate, maintaining accuracy is dominated by the analog signal chain settles *inside* the PWM/SVM sampling window. The key requirement is the window budget $t_{\text{req}} = t_{\text{acq}} + t_{\text{blank}} + t_{\text{margin}}$; conversion time t_{conv} (MSPS) is secondary because the S&H can convert after a settled capture.

Modulation boundaries from settling time. The modulation headroom is set by t_{req} and T_{sw} , not by ADC “bandwidth”:

- **Dual-shunt:** no timing-driven *lower* bound on m . The *upper* bound ensuring a valid per-phase plateau is

$$m_{\text{max}} = 1 - \frac{2t_{\text{req}}}{T_{\text{sw}}}, \quad d_{\text{max}} = 1 - \frac{t_{\text{req}}}{T_{\text{sw}}}, \quad d_{\text{min}} = \frac{t_{\text{req}}}{T_{\text{sw}}}.$$

With the practical chain (150 ns ADC + 600 ns op-amp, $t_{\text{req}} \approx 0.75$ – $1.05 \mu\text{s}$ at 20 kHz), this supports $m \approx 0.96$ – 0.97 and $\sim 98\%$ per-phase duty without PWM shaping.

- **Single-shunt:** Determining the same modulation boundaries equation for single shunt use case is more complex and out of the scope for this paper.

Impact on efficiency and topology choice. Because of current error in sensorless FOC also drives angle error, meeting the timing condition $t_{\text{acq}} \leq t_{\text{win}}$ directly limits torque misalignment and the associated copper-loss increase. In practice, dual- and three-shunt topologies offer wide windows and simpler timing, while single-shunt remains the most cost- and pin-efficient but is *window-limited* and sensitive to t_{req} .

Future work

A companion paper will (i) derive minimum system ENOB targets from application-level ϵ_θ via measured $S_{\theta I}$, (ii) formalize modulation index boundaries for single-shunt, and (iii) present hardware validation of t_{req} under PWM-induced transients.

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